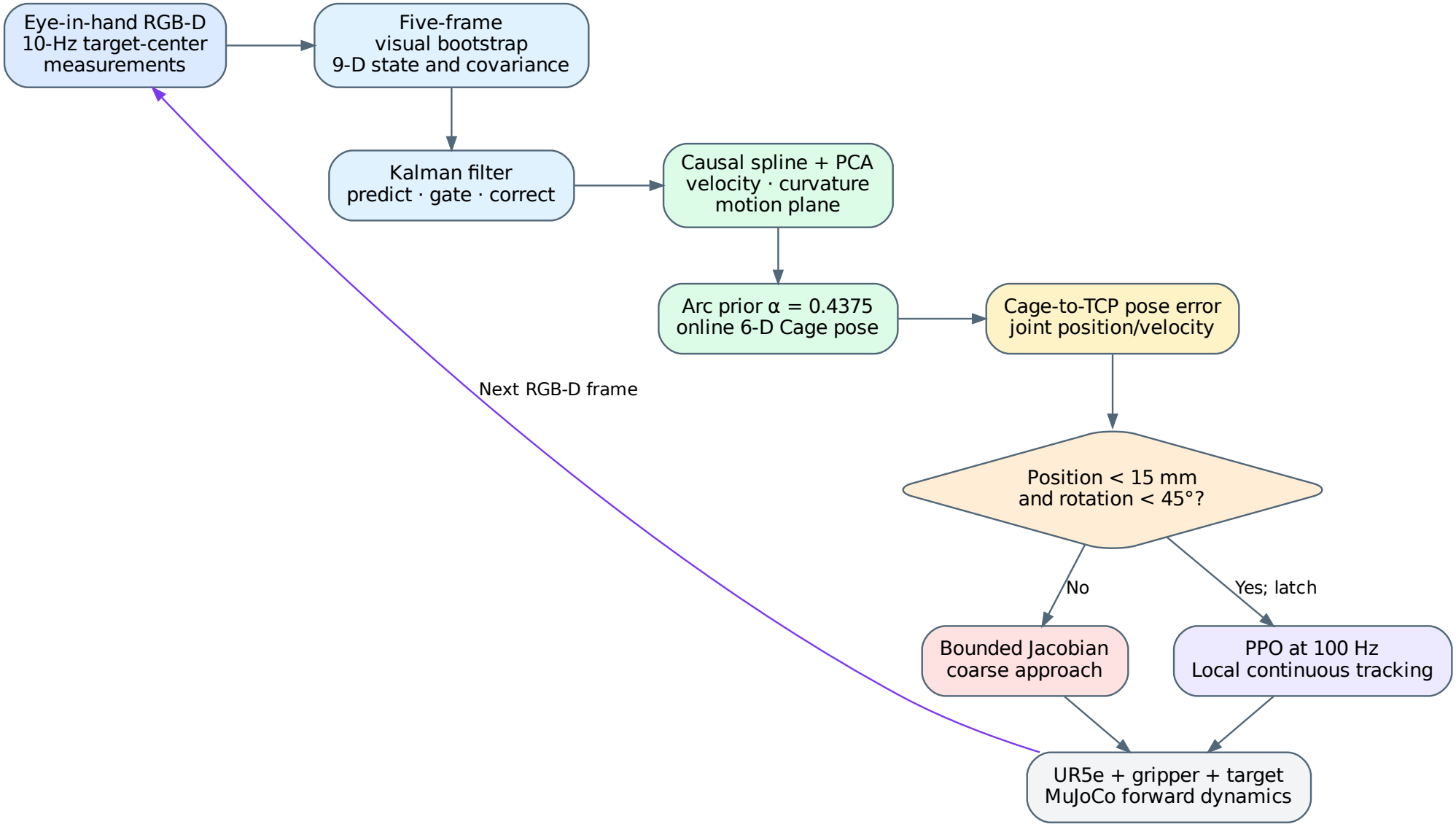
Eye-in-hand RGB-D
10-Hz target-center
measurements
Five-frame
visual bootstrap
9-D state and covariance
Kalman filter
predict · gate · correct
Causal spline + PCA
velocity · curvature
motion plane
Arc prior α = 0.4375
online 6-D Cage pose
Cage-to-TCP pose error
joint position/velocity
Position < 15 mm
and rotation < 45°?
No
Yes; latch
Bounded Jacobian
coarse approach
PPO at 100 Hz
Local continuous tracking
UR5e + gripper + target
MuJoCo forward dynamics
Next RGB-D frame